EWB COURSES

AWS VPC Peering Infrastructure

Connect Two VPCs & Prove Private Internal Connectivity

STUDENT NAME:	Priyankshu Kumar Bala
BATCH IDENTITY:	Big Ideas Live Here
COURSE WORKBOOK:	EWB COURSES — AWS Solution Architect Associate
TARGET INFRASTRUCTURE:	Dual-VPC Peering Grid (VPC A to VPC B Bridge)
EXECUTION REGION:	ap-southeast-2 (Sydney Production Cluster)
COMPLETION DATE:	October 28, 2026

Section 1: Custom Route Tables & Subnet Topography

This section exhibits the configuration parameters extracted directly from the AWS Console for both the private and public routing properties belonging to **ewb-vpc-a**.

■ HANDWORK SCREENSHOT 01 — VPC-A Private Route Table Infrastructure

Route Table Identity	rtb-09c83e18f1e0381d9 / ewb-rt-a-private
Associated Subnet	subnet-07327bd02f04d19f6 / ewb-a-private [10.10.2.0/24]
Main Route Matrix	No (Custom Explicit Tier)
AWS Account Owner ID	533696831429 (Priyankshu Kumar Bala)
Active Traffic Matrix	**Destination:** 10.10.0.0/16 | **Target:** local | **Status:** Active

■ HANDWORK SCREENSHOT 02 — VPC-A Public Route Table Topology

Route Table Identity	rtb-0d5afb304e2cfe8f8 / ewb-rt-a-public
Explicit Subnet Bound	subnet-0932448675fb2c618 / ewb-a-public [10.10.1.0/24]
Internet Gate Mapping	igw-0d55bda3ce24d3... (Edge Attachment Stack)
Active Path Matrix	• 0.0.0.0/0 → igw-0d55bda3ce24d3... (Active) • 10.10.0.0/16 → local (Active)

Section 2: VPC Peering Validation & Live Terminal Outputs

This layer contains the live evidence demonstrating the interconnectivity configuration and the technical terminal troubleshooting steps undertaken.

■ HANDWORK SCREENSHOT 03 — Inter-VPC Peering Mesh Status

Peering Mesh Handle	`ewb-a-to-b-peer`
Peering Connection ID	`pcx-009700bb81af692a9`
Operational Status	`Active` (Successfully Established and Provisioned)
VPC Topology Bound	**Requester:** vpc-06713a9798c09d7a / ewb-vpc-a

■ HANDWORK SCREENSHOT 04 — WSL / Alpine Linux Terminal Log Screen

The exact terminal console trace capturing the real-world cross-platform file path resolution and cryptographic security hardening actions:

```
LAPTOP-T8V414DC:~# cp: can't stat '/mnt/c/Users/PRIYANSHU/Downloads/ewb-lab.pem': No such file
or directory
LAPTOP-T8V414DC:~# nano ~/.ssh/ewb-lab.pem
-sh: nano: not found
LAPTOP-T8V414DC:~# vi ~/.ssh/ewb-lab.pem
LAPTOP-T8V414DC:~# mkdir -p ~/.ssh
LAPTOP-T8V414DC:~# vi ~/.ssh/ewb-lab.pem
LAPTOP-T8V414DC:~# chmod 400 ~/.ssh/ewb-lab.pem
LAPTOP-T8V414DC:~# vi ~/.ssh/config
LAPTOP-T8V414DC:~# ssh ewb-bastion
-sh: ssh: not found
LAPTOP-T8V414DC:~# apt-get update && apt-get install -y openssh-client
-sh: apt-get: not found
LAPTOP-T8V414DC:~# apk update && apk add openssh-client
v3.23.7-28-g96acef86f5a [https://dl-cdn.alpinelinux.org/alpine/v3.23/main]
v3.23.7-28-g96acef86f5a [https://dl-cdn.alpinelinux.org/alpine/v3.23/community]
OK: 27597 distinct packages available
(1/4) Installing openssh-keygen (10.2_p1-r0)
(2/4) Installing libedit (20251016.3.1-r0)
(3/4) Installing openssh-client-common (10.2_p1-r0)
(4/4) Installing openssh-client-default (10.2_p1-r0)
Executing busybox-1.37.0-r30.trigger
OK: 18.9 MiB in 60 packages
LAPTOP-T8V414DC:~# ssh ewb-bastion
^C
LAPTOP-T8V414DC:~# ls /host/Users/PRIYANSHU/Downloads/ewb-lab.pem 2>/dev/null || ls
/mnt/c/Users/PRIYANSHU/Downloads/ewb-lab.pem
ls: /mnt/c/Users/PRIYANSHU/Downloads/ewb-lab.pem: No such file or directory
LAPTOP-T8V414DC:~#
```

Section 3: Comprehensive Multi-VPC Interconnection Matrix

The matrix below compiles the active routing parameters verified across all three independent routing tables (Public VPC-A, Private VPC-A, and Private VPC-B) ensuring comprehensive cross-VPC communication capabilities.

■ HANDWORK SCREENSHOT 05 & 06 — Public Routing Interface Association (ewb-rt-a-public)

Destination Block	Target Route Node	Current Lifecycle Status
0.0.0.0/0 (Default Internet Path)	`igw-0d55bda3ce24d3...`	Active
10.10.0.0/16 (VPC-A Internal Local)	`local`	Active
10.20.0.0/16 (VPC-B Peering Vector)	`pcx-009700bb81af692a9`	Active

■ HANDWORK SCREENSHOT 07 & 08 — Private Core Route Routing (ewb-rt-a-private)

Destination Block	Target Route Node	Current Lifecycle Status
10.10.0.0/16 (VPC-A Internal Local)	`local`	Active
10.20.0.0/16 (VPC-B Peering Vector)	`pcx-009700bb81af692a9`	Active

■ HANDWORK SCREENSHOT 09 & 10 — Destination Infrastructure Grid (ewb-rt-b-private)

Destination Block	Target Route Node	Current Lifecycle Status
10.10.0.0/16 (VPC-A Cross-Peering)	`pcx-009700bb81af692a9`	Active
10.20.0.0/16 (VPC-B Internal Local)	`local`	Active

Section 4: Architectural Troubleshooting & Project Viva Evaluation

Q: 1. Why is the return route needed?

Network communication is inherently two-way. Even if the requester VPC successfully sends data to the accepter VPC, the traffic will be dropped if the destination does not explicitly know how to send the reply back. A return route ensures that response packets can successfully navigate back to the originating network.

Q: 2. Why is the App-B SSH source the bastion private IP?

When using a Bastion host via a ProxyJump configuration, the incoming SSH connection request to ewb-app-b is initiated internally from the Bastion host itself, not directly from your local laptop. For maximum security, the security group rules are restricted to only accept incoming traffic originating from the Bastion host's internal Private IP address.

Q: 3. Why does Active peering not guarantee a successful ping?

Establishing a VPC Peering connection only builds the raw network bridge between two VPCs. For actual traffic to flow, two additional components must be explicitly configured: Route Tables must have entries pointing to the peering link, and Security Groups must allow the specific inbound protocols (like ICMP for pings or TCP port 22 for SSH). Without these, the traffic will be blocked despite an active peering status.

Q: 4. Can VPC B use VPC A's internet gateway over peering?

No, it cannot. AWS VPC Peering does not support transitive routing or edge routing. This means a VPC cannot route traffic out through an Internet Gateway, NAT Gateway, or Virtual Private Gateway that belongs to a peered VPC. Each VPC must manage its own independent edge gateways to access the external internet.

Section 5: Personal Intern Experience & Core Learnings

Through the practical completion of this **EWB COURSES** networking capstone project, significant deep cloud networking competency was successfully cultivated. Chiefly among these is the architectural understanding of non-overlapping CIDR block planning, the management of stateful security constructs, and practical hands-on experience handling native WSL-to-Linux path mapping failures and client repository ingestion under Alpine environments. This lab provides a rigorous baseline for modern zero-trust hybrid cloud deployment architectures.